

UBT410: PLANT BIOTECHNOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: The course will enable the students to acquire knowledge about various techniques like micropropagation, single cell culture, suspension culture, protoplast culture, hairy root culture and various techniques of recombinant DNA technology to produce genetically modified organisms with novel characters.

Syllabus

Introduction, Aim and Scope of Plant Biotechnology: Major challenges and prospects of traditional and modern plant biotechnology, Important milestones of plant biotechnology

Plant Tissue Culture and protoplast culture—its history, development and applications, Plant tissue culture media, Plant growth regulators, Types of cultures, Protoplast culture and somatic hybridization.

Micropropagation: Techniques and various steps involved in micropropagation, Production of disease-free plants and certification of tissue culture raised plants.

Production of Haploid Plants: Androgenesis and Gynogenesis, Significance and uses of haploids. Embryo culture and embryo rescue and its applications in plant improvement.

Strategies for Producing Novel Plants: Manipulation of Phenotypic Traits: Strategies of molecular cloning of plant genes, direct and indirect gene transfer methods, rDNA approaches for introducing herbicide tolerance, pest resistance, plant disease resistance, Abiotic & biotic stress tolerance, various strategies for the improvement of crop yield and quality, Applications of plant transformations/ transgenics, Molecular farming of commercially/pharmaceutically important products.

Secondary Metabolite Extraction: Primary vs secondary metabolites, Role of plant tissue culture in secondary metabolite production, Hairy root culture, Strategies for the enhancement of secondary metabolite production.

Germplasm Conservation and Somaclonal Variations: Isolation of somaclonal variants, Applications and limitations of somaclonal variations, Gametoclonal variations. Germplasm and methods of its conservation and Cryopreservation

Self-Learning: Transgenics-Issues and Concerns, Biosafety, Societal and ethical concerns on genetically modified foods and crops.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Course Learning Objectives (CLO)

The students will be able to:

1. Familiarize with organization of PTC Lab., aseptic manipulations and learn techniques of culturing tissues, single cells, protoplast and anther culture, hairy root culture and germplasm conservation.
2. Undertake large scale in vitro propagation of plants through micropropagation.
3. Generate plants with desirable/novel traits through genetic manipulations using different methods of gene transfer and marker associated selections.
4. Recognize the importance of plant secondary metabolites, their commercial production.

Text Books

1. Slater, A., Scott, N.W., and Fowler, M.R., Plant Biotechnology, Oxford University Press (2008).
2. Chawala, HS, Introduction to Plant Biotechnology, Third Edition, Oxford & IBH (2017).
3. Primrose, S.B. and Twyman, R.M., Principles of Gene Manipulation and Genomics, 7th Edition, Blackwell Publishing (2006).

Reference Books

1. Satyanarayana, U., Yeast Biotechnology: Diversity and Applications, Springer (2009).
2. Razdan, M.K., Introduction to Plant Tissue Culture, Science Publishers (2003).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25-35
2	EST	35-45
3	Sessionals (May include Assignments/Quizes)	30

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